



IoT-Enabled Automated Traffic Lights Control System

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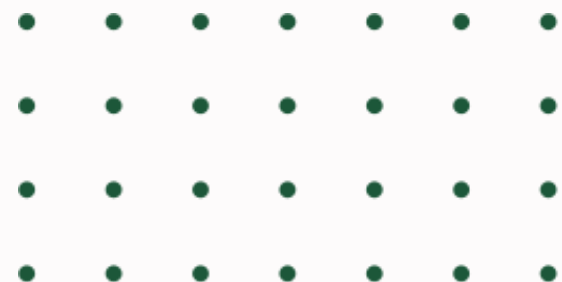
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OVERVIEW

Abstract

This project proposes an IoT-based smart traffic light system using IR/ultrasonic sensors and an ESP32 to dynamically adjust signal timings based on real-time vehicle density. It aims to reduce wait times, enhance safety, and improve urban traffic flow, featuring an LCD for monitoring and optional IoT connectivity.

Urban traffic congestion, worsened by fixed signal timings, leads to delays and pollution. Our project proposes an IoT-based smart traffic system using sensors and automation to adaptively control traffic lights in real-time, improving flow and reducing issues.

Introduction

PROBLEM STATEMENT

Current traffic lights are often ineffective in managing dynamic traffic volumes. This results in long queues, fuel wastage, and increased CO₂ emissions. Manual intervention or preset timers fail to optimize flow, especially during peak hours or emergencies.

OBJECTIVE

- To develop an IoT-based system that detects vehicle density at intersections.
- To automate traffic light control using sensor data for optimized flow.
- To reduce wait time and improve traffic efficiency.
- To allow remote monitoring and manual override when required.
- To ensure energy efficiency and low operational cost.

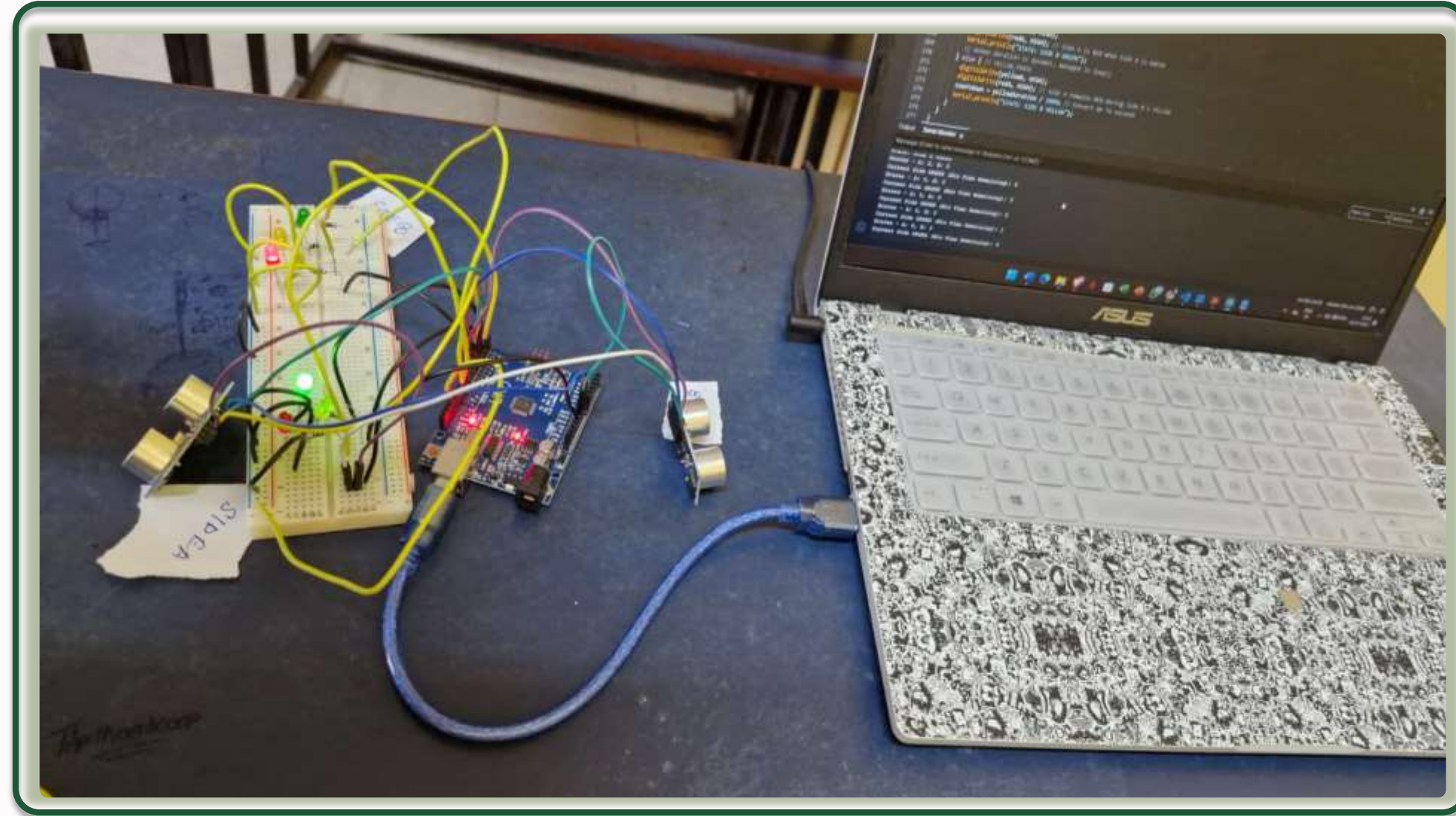


METHODOLOGY

- **COMPONENT SELECTION AND PURPOSE**

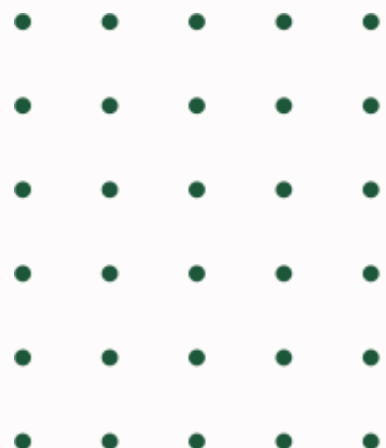
COMPONENT	SPECIFICATION	PURPOSE
Arduino UNO	Atmega328P based microcontroller	Central controller for processing data and signal timing.
Ultrasonic Sensors	HC-SR04	Measures distance to count vehicles on each lane.
3-Color LEDs (Red, Yellow, Green)	Standard 5mm LEDs	Simulates traffic lights.
Jumper Wires & Breadboard	-	Used for prototyping connections.
Power Supply	5V/12V DC	Powers Arduino UNO and sensors.

SYSTEM DESIGN



- Each lane is equipped with an ultrasonic sensor.
- Sensor data is collected by the Arduino UNO.
- Based on the number of vehicles detected, signal timing is adjusted.

- Priority can be assigned for emergency lanes.
 - LEDs simulate traffic light phases in prototype.
 - Data is displayed on monitor screen and optionally sent to a cloud platform.

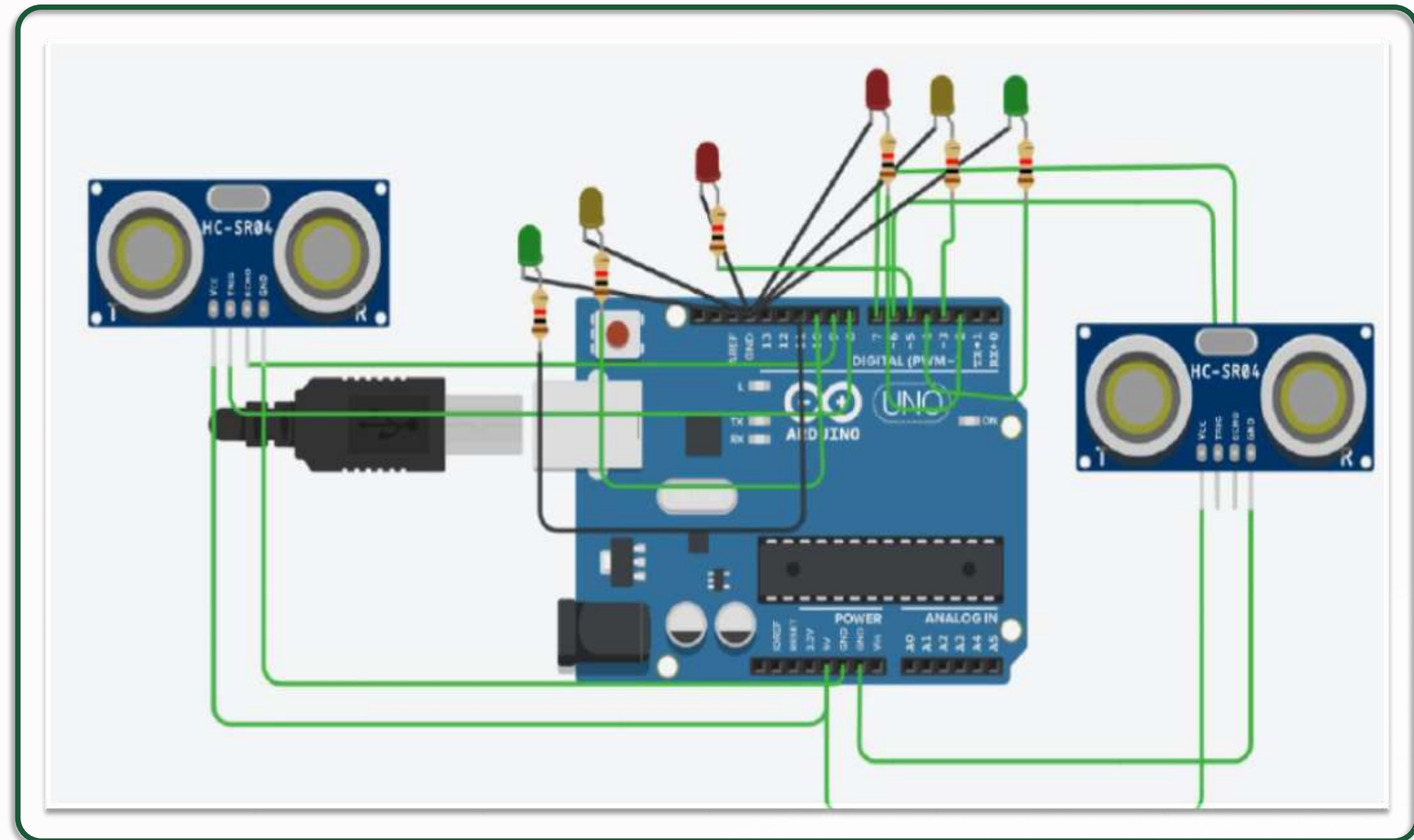


Working Model



CIRCUIT

- **Arduino UNO is programmed via Arduino IDE.**
- **Real-time sensor readings influence LED switching logic.**
- **Delay timers are dynamically adjusted based on traffic load.**
- **Blynk or Firebase integration enables live data viewing.**



DISCUSSION

TESTING AND OPTIMIZATION

The system was tested under various indoor conditions. Threshold values were fine-tuned for sensitivity and energy efficiency. User experience, response time, and system stability were evaluated to ensure reliable operation.

FINAL INTEGRATION

All components were neatly placed inside the enclosure. Power management and cable routing were finalized. The system operated autonomously, with the option for manual overrides, making it ready for real-world use.



APPLICATIONS

1

Urban Intersections: Smartly manages dense traffic, especially during peak hours.

2

Emergency Routes: Prioritizes ambulance or firetruck movement.

3

Campus Roads: For internal smart traffic control in universities or corporate zones

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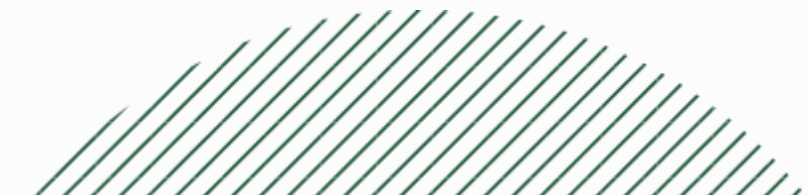
Smart Cities: Integrates with broader urban management systems.

5

Developing Regions: Cost-effective solution for local authorities.



Future scope

- **AI-Based Prediction:** Use machine learning to anticipate congestion trends.
 - **Camera-Based Monitoring:** Replace sensors with image processing for accurate detection.
 - **V2X Communication:** Enable vehicles to communicate directly with the system.
 - **Mobile App Control:** For emergency services or traffic police override.
 - **Solar-Powered Units:** Deploy standalone units in remote or power-constrained areas.
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CONCLUSION

This IoT-based automated traffic light system presents an efficient, scalable, and cost-effective solution for modern traffic challenges. By using real-time vehicle detection and dynamic signal control, it significantly improves traffic flow and reduces wait time and pollution. The system is easily upgradable and suitable for smart city deployments. It aligns with UN SDGs on sustainable cities, climate action, and innovation.

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A dark green geometric shape, resembling a stylized arrow or a corner, is located in the top-left corner of the image. It has a thick, dark green outer edge and a slightly lighter green inner edge, creating a layered effect.

**THANK
YOU!**